# Vamsi Vajja

**Senior AI/ML Engineer**

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## Professional Summary

I am a Senior AI/ML Engineer with over 8+ years of experience designing and delivering scalable, production grade data and machine learning systems across domains such as e-commerce, mobility, retail, and financial services. I have led the development of intelligent platforms that transform large scale, real time data into actionable insights and automated decision making systems. My expertise spans the full ML lifecycle from data ingestion and feature engineering to model training, deployment, and monitoring leveraging Python, SQL, PyTorch, TensorFlow, and Scikit learn. I have built robust data pipelines and real-time processing systems using Apache Kafka, Celerys and Redis, ensuring high throughput and low latency performance in distributed environments. I have also developed enterprise grade AI systems using Retrieval Augmented Generation (RAG), Large Language Models (LLMs), LangChain, and vector databases like Pinecone and OpenSearch, enabling context aware, compliant knowledge assistants. My work includes building validation engines with rule based systems, implementing pricing and content guardrails, and ensuring data quality at scale using optimized queries in MySQL and caching strategies like Memcached. From a systems perspective, I specialize in designing cloud native and microservices based architectures using AWS (Lambda, S3, CloudWatch, API Gateway), Docker, Kubernetes, and FastAPI/Node.js for scalable backend services. I have implemented MLOps pipelines using MLflow, CI/CD tools like Jenkins and Azure DevOps, and ensured observability through logging and monitoring frameworks. My ability to bridge data science and engineering allows me to build highly reliable, efficient, and production ready AI systems.

## Skills

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| **GenAI & LLMs** | Agentic AI, RAG, LLM-as-a-Judge, Fine-tuning, Prompt Engineering, LangChain, LlamaIndex, vLLM, DeepSeek, OpenAI API |
| **Machine Learning** | PyTorch, TensorFlow, Hugging Face, Scikit-learn, XGBoost, LightGBM, Sentence Transformers, Deep Learning |
| **Data Platforms** | Databricks (Delta Lake, Unity Catalog), Spark, PySpark, Kafka, Airflow, Flink, Exactly Once Semantics |
| **Cloud & Infrastructure** | AWS Bedrock, GCP Vertex AI and BigQuery, Azure OpenAI, Cognitive Search, Docker, Kubernetes, Terraform |
| **MLOps & Monitoring** | MLflow, DVC, LangSmith, Weights & Biases, Arize AI, Fiddler AI, Prometheus, Grafana, GitHub Actions |
| **Databases** | Vector DBs (OpenSearch, FAISS, Chroma), PostgreSQL, MongoDB, DynamoDB, Redis Streams |
| **Languages** | Python, SQL, C, Java, Bash, JavaScript |
| **API & Security** | FastAPI, Flask, OAuth2, JWT, PII Redaction, MCP, FastMCP |

## Work Experience

**Barclays** | Whippany, NJ — Jan 2024 – Present
*Senior AI/ML Engineer*

**Overview:** Barclays US Consumer Bank identified manual After Call Work (ACW) as a key bottleneck in contact center operations, where agents spent significant time summarizing every customer interaction post call. This project leveraged Generative AI to automatically capture call intent, agent actions, compliance flags, and next steps from raw transcripts, eliminating manual effort and enabling agents to focus entirely on customer experience.

- Developed an enterprise grade GenAI knowledge assistant using Python, FastAPI, LangChain, and LangGraph, enabling users to query financial policies and compliance documents using natural language. Designed the system for scalable deployment on AWS to support high performance enterprise usage.

- Implemented a Retrieval Augmented Generation (RAG) pipeline to ground LLM responses in enterprise knowledge sources, integrating with Amazon Bedrock for foundation models. Ensured context aware, reliable outputs while minimizing hallucinations in domain-specific queries.
- Deployed a vector based semantic search system by generating embeddings and indexing them in Amazon OpenSearch and FAISS, enabling efficient similarity search with sub second (<500ms) retrieval latency.
- Built scalable document ingestion and preprocessing pipelines to extract and process unstructured data from PDFs and internal repositories, storing raw documents in Amazon S3. Applied intelligent chunking strategies to preserve semantic context and improve retrieval quality.
- Implemented advanced document chunking techniques using LangChain text splitters, including recursive and fixed size chunking with overlap. Optimized chunking strategies to improve contextual grounding and increase retrieval relevance by 30–35%.
- Integrated semantic retrieval with Top-k search and re-ranking over vector databases enabling accurate identification of relevant document segments and improving answer precision compared to traditional keyword-based approaches.
- Designed and implemented agentic AI workflows using LangGraph, enabling multi step reasoning, dynamic decision making, and tool invocation (retrieval, summarization, compliance checks) to autonomously generate context aware responses for enterprise use cases.
- Optimized system performance by tuning embedding generation, vector indexing, and prompt strategies, improving response consistency and first response correctness by 25% through iterative evaluation.
- Built secure and scalable RESTful APIs using FastAPI, deployed on Amazon EC2/EKS, and exposed via Amazon API Gateway to support real time query handling and seamless integration with enterprise systems.
- Designed and implemented evaluation and feedback pipelines using curated financial queries to benchmark retrieval and generation quality, enabling continuous improvement in system performance.
- Implemented query caching and embedding reuse strategies to reduce redundant computations and improve system efficiency, optimizing performance under repeated query workloads.
- Integrated monitoring and logging pipelines using Amazon CloudWatch to track performance, query latency, and retrieval accuracy, enabling proactive debugging and ensuring high availability of GenAI workflows.

**Technical Stack:** AWS Bedrock, LangChain, LangGraph, RAG, Amazon OpenSearch, FAISS, Amazon S3, AWS Lambda, API Gateway, AWS IAM, Amazon CloudWatch, Python (FastAPI, Pydantic), Docker, Amazon EKS, Amazon EC2, AWS Application Load Balancer, AWS VPC, AWS CloudFormation, Boto3, Redis.

**Target** | Brooklyn Park, MN Aug 2022 – Jul 2023
*Machine Learning Engineer*

**Overview:** Target's retail operations depend on moving billions of units of inventory across 2,000 stores with surgical precision. This project focused on building a centralized Machine Learning platform to replace legacy statistical forecasting. By integrating real time data such as local weather, social trends, and live clickstream activity, the system predicts SKU level demand with high accuracy.

- Designed a centralized ML platform for demand forecasting and personalization using Python, PyTorch, TensorFlow, and XGBoost, replacing legacy statistical models and improving prediction consistency.
- Built deep learning recommendation models using PyTorch and TensorFlow, improving CTR by 15% through customer segmentation, embedding techniques, and behavioral feature engineering.
- Architected end-to-end MLOps pipelines on GCP leveraging Vertex AI, Kubeflow, and MLflow, automating data ingestion, model training, validation, and deployment workflows across environments.
- Developed scalable model serving APIs using FastAPI, Docker, and Kubernetes (GKE), handling millions of requests with sub-100ms latency during peak events like Black Friday.
- Processed multi 'terabyte datasets using PySpark and BigQuery to build real-time and batch feature pipelines with Redis, reducing feature engineering time by 40% and enabling low-latency inference.
- Implemented model monitoring with data validation and drift detection pipelines, triggering automated retraining workflows to maintain prediction accuracy in production systems.
- Built distributed training workflows on Vertex AI using GPU acceleration, reducing model training time by 30% and improving experimentation speed for forecasting models.
- Established CI/CD and automated infrastructure using GitHub Actions and Terraform alongside GKE auto scaling and multi model endpoints, reducing operational costs by 25% while maintaining high availability.

- Designed A/B testing frameworks to evaluate model variants in production, using statistical analysis and controlled experiments to guide feature rollouts and model selection.
- Collaborated with cross functional teams including Product and Supply Chain to translate business requirements into ML objectives, aligning model outputs with inventory and demand goals.

**Technical Stack:** Python, PyTorch, TensorFlow, XGBoost, GCP (Vertex AI, BigQuery, GKE), Kubeflow, MLflow, PySpark, Redis, FastAPI, Docker, Kubernetes, Terraform, GitHub Actions.

**Uber** | United States May 2019 – Nov 2021
*Senior Data Engineer*

**Overview:** Developed a real time data platform at Uber to process high volume ride demand, driver availability, and geolocation signals for dynamic pricing and dispatch optimization. Built scalable streaming and batch pipelines using Kafka and Spark to compute demand-supply imbalances and generate reliable forecasting features. Integrated these pipelines with downstream pricing and ML systems to enable accurate surge pricing and efficient ride allocation. This significantly improved system responsiveness, reduced rider wait times, and enhanced overall platform efficiency under peak traffic conditions.

- Engineered a real time data pipeline using Apache Kafka and Apache Spark Streaming to process millions of ride events, enabling dynamic demand supply estimation across geo regions with sub minute latency.
- Designed scalable data models on Amazon S3, Apache Hive, and Presto, optimizing partitioning and query performance to support high throughput analytical workloads for pricing and dispatch systems.
- Built batch/streaming feature pipelines via PySpark and Apache Airflow, integrating them with Michelangelo (Uber's ML platform) to deliver curated datasets and improve real time pricing accuracy.
- Optimized end to end pipeline performance and data quality by tuning Spark jobs and Kafka consumers while implementing rigorous validation checks to ensure reliable inputs for critical decision systems.

**Technical Stack:** Apache Kafka, Apache Spark (PySpark, Spark Streaming), Amazon S3, Apache Hive, Presto, Apache Airflow, Michelangelo, SQL, Python, Hadoop Ecosystem.

**Flipkart** | Bengaluru, India Mar 2017 – Apr 2019
*Python Developer*

**Overview:** Flipkart's marketplace growth depended on onboarding thousands of new sellers rapidly without compromising catalog quality. This project focused on building a scalable self-serve onboarding platform that automated product listing, validation, and publishing workflows. By replacing manual moderation with intelligent validation pipelines, the system significantly reduced onboarding time while maintaining strict quality standards for pricing, images, and product metadata.

- Designed and implemented a scalable self-serve seller onboarding platform using Python (Django), enabling independent product publishing and reducing manual onboarding efforts by 60%.
- Architected asynchronous processing pipelines and validation engines using Celery, Redis, RabbitMQ, and Pillow to automate image processing and enforce rule-based pricing guardrails for 50k+ daily uploads.
- Built and exposed RESTful APIs for seamless integration between the Seller Portal, Order Management System (OMS), and logistics services, driving real-time data synchronization.
- Slashed dashboard latency by 30% via high-frequency MySQL query optimization and Memcached, while ensuring system reliability through 90%+ unit test coverage using PyTest.

**Technical Stack:** Python (Django/Flask), Celery, Redis, MySQL, RabbitMQ, Pillow, PyTest, Docker, Git.

## Certifications

- IBM - Introduction to Artificial Intelligence
- University of Washington - Machine Learning Foundations
- IBM - Introduction to Big Data with Spark and Hadoop
- Duke University - Cloud Computing Foundations

## Education

**University of Maryland, Baltimore County**
*Master of Science in Computer Science*